

Reason from shape properties

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth. Copy or complete the first step.

2. Angles around a point are 85 degrees, 140 degrees and x . Find x .

3. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth. Copy or complete the first step.
Answer 75 degrees.
2. Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer 135 degrees.
3. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees.
4. Miro says: Miro estimates from the drawing even when the diagram is not to scale. Is this correct? Explain.
Answer Use labelled values and known geometric facts, then check the total angle relationship.

Reason from shape properties

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

6. Check this result using an inverse, estimate or known fact: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

2. Angles around a point are 85 degrees, 140 degrees and x . Find x .

7. Prove or explain your reasoning: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

3. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.

4. Solve this independently and show every stage: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Angles around a point are 85 degrees, 140 degrees and x . Find x .

10. Write and solve a similar question to: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

Teacher answers and checking prompts

1. A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer 75 degrees.
2. Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer 135 degrees.
3. One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer 68 degrees.
4. Solve this independently and show every stage: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer 75 degrees.
5. Use a second method or representation for: Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer Expected result: 135 degrees. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer Expected result: 68 degrees. A valid check must be shown.
7. Prove or explain your reasoning: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees. The explanation must show why the method works.
8. Identify and correct this misconception: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason from shape properties

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: A triangle has angles 47 degrees and 68 degrees. Find the third angle.

6. Create a new example that tests the same idea as: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.

2. Prove the answer to this question, rather than only calculating it: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Angles around a point are 85 degrees, 140 degrees and x . Find x .

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 68 degrees.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: A triangle has angles 47 degrees and 68 degrees. Find the third angle.
Answer 65 degrees. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A quadrilateral has three angles of 92 degrees, 88 degrees and 105 degrees. Find the fourth.
Answer Expected result: 75 degrees. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Angles around a point are 85 degrees, 140 degrees and x . Find x .
Answer Expected result: 135 degrees. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 68 degrees.
Answer One valid reconstruction is: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro estimates from the drawing even when the diagram is not to scale.
Answer Use labelled values and known geometric facts, then check the total angle relationship.
6. Create a new example that tests the same idea as: One angle where two straight lines cross is 68 degrees. Find the vertically opposite angle.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.